DIPESH MARATHE

Gen AI and Full-Stack Developer | LLM Systems, RAG and MCP
+91 8766726443 | dipeshmarathe9765@gmail.com | linkedin.com/in/dipesh9765 | github.com/dipesh9765 | leetcode.com/dipesh9765

PROFESSIONAL SUMMARY

Gen AI and full-stack developer with over two years of experience shipping production AI features at Novatech Software: multi-provider LLM integrations (OpenAI, Azure OpenAI), a RAG pipeline on user owned data (+87 percent user satisfaction), a custom Model Context Protocol (MCP) server with agentic workflows, and an AI Governor Server that rate-limits, queues, budgets and audits every LLM call. Builds across React and Sencha ExtJS front ends and Node.js (Express 5), Python and Java services on PostgreSQL, pgvector and Redis. Strong foundation in DSA and system design: 1,000+ problems solved, CodeChef rating 1630. CGPA 8.96.

TECHNICAL SKILLS

AI and Agentic: RAG, OpenAI API, Azure OpenAI, Agentic Workflows, Model Context Protocol (MCP), Custom MCP Servers, Prompt Engineering (Strict JSON Schemas), AI Governance
Languages: JavaScript (ES6+), TypeScript, Python, Java, C/C++, SQL
Backend: Node.js, Express.js (Express 5), REST APIs, Microservices
Frontend: React, Sencha ExtJS, Tailwind CSS, HTML/CSS
Databases: PostgreSQL, pgvector, MySQL, Redis
Testing and Security: Jest, Unit Testing, Postman, Peer Code Review, Input Validation, Rate Limiting, Audit Logging
Tools and Core CS: Git, Jupyter, VS Code, Agile / SCRUM, DSA, OOP, System Design, Distributed Systems, Concurrency

PROFESSIONAL EXPERIENCE

Associate Software Developer | Novatech Software, Nagpur, MH | Jan 2024 - Present
LLM and AI Integration (May 2024 - Present)

- Engineered an AI Governor Server (Express.js / Express 5, Redis, PostgreSQL) managing LLM API handshakes: rate limiting, request queuing, token budgeting, and audit logging.
- Implemented a Retrieval Augmented Generation (RAG) pipeline on user owned data, improving AI answer quality and raising user satisfaction by 87 percent.
- Architected multi-provider LLM API integrations (OpenAI, Azure OpenAI) into a production AI assistant, increasing user productivity by 68 percent.
- Built a custom MCP server for the company's product requirements management tool, encoding its own configuration and business rules so LLMs can securely connect with the product's APIs and business systems; orchestrated agentic workflows on top of it.
- Designed prompt pipelines enforcing strict JSON schema outputs for reliable multi-step tasks.
- Designed AI governance for the free-tier AI features of the product, controlling and auditing usage of AI capabilities offered to users.
- Built LLM powered record auto generation and quality analysis, reducing manual data entry and errors.
- Wrote Jest unit tests for the usage tracking and cost calculation module, verifying enforcement of AI usage limits.

Backend and Full-Stack Engineering (May 2024 - Present)

- Built a Use Case Flow Editor end to end: flow authoring, UML activity diagram auto generation, and AI driven test case synthesis.
- Shipped a role-based Gridstack dashboard with configurable widgets, improving team workflow efficiency by 73 percent.
- Built custom Sencha ExtJS components (syntax highlighters, rich data entry widgets) in a complex data pipeline UI.
- Led migration of legacy monolithic desktop tools to a scalable Node.js web stack, decoupling the front end from backend logic.
- Applied secure coding practices (input validation, sensitive data handling), including for banking and financial services customers.
- Collaborated with product managers and designers on requirements and solution strategy; participated in peer code reviews.
- Leading mobile app development for remote dashboard access and distributed test execution.

Engineering Internship (Jan 2024 - Apr 2024)

- Built a content parsing engine reducing record creation time by 95 percent.
- Fixed critical cross-browser bugs; prototyped three features that shipped to production.

PROJECTS

KitePulse-AI, Agentic Trading Assistant Node.js, Express 5, Zerodha Kite API, Telegram API, MCP

- Built an autonomous market-monitoring agent with conditional alerts and rate limiting, timeouts, and retries.
- Exposed trading state to the LLM through MCP.

AI Calorie and Nutrition Tracker TypeScript, Express 5, LLM structured outputs, pgvector

- Built an agent that logs nutrition from unstructured input using strict JSON schemas.
- Used pgvector retrieval of past history, with human approval before any state change.

Big Data Distributed Search Engine Python, MySQL, JavaScript

- Built keyword search over 10 million plus records using inverted indexing and concurrent processing, achieving a 70 percent reduction in latency.

EDUCATION

B.Tech, Computer Science Engineering | Dec 2020 - Jun 2024
Government College of Engineering, Amravati | CGPA: 8.96 / 10

ACHIEVEMENTS AND RECOGNITION

- Smart India Hackathon 2022 Finalist, ranked in the top 1 percent of over 50,000 teams nationwide.
- Bright Beginner Award 2025, company recognition for standout impact within the first six months.
- Talent Hunt 2024 Winner (L&T EduTech), ranked first among over 10,000 participants.
- DSA and Competitive Programming Advisor, Google Developer Student Club, mentored over 100 students.
- Solved over 1,000 problems across LeetCode, GeeksforGeeks, and CodeChef, with a CodeChef rating of 1630.